

## Week-1

Program-1:

```
GNU nano 8.7.1
#include <stdio.h>
int main(){
printf("Welcome to operating Systems\n");
return 0;
}
```

Output:

```
akhil@Akhil:~/OS$ gcc hello.c -o hello
akhil@Akhil:~/OS$ ./hello
Welcome to operating Systems
```

Program-2:

```
GNU nano 8.7.1
#include<stdio.h>
int main(){
char name[30];
printf("Enter your name: ");
scanf("%s",name);
printf("Welcome %s\n",name);
return 0;
}
```

Output:

```
akhil@Akhil:~/OS$ gcc name.c -o name
akhil@Akhil:~/OS$ ./name
Enter your name: akhil
Welcome akhil
```

Program-3:

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
int main(){
printf("Process ID=%d\n",getpid());
return 0;
}
```

Output:

```
akhil@Akhil:~/OS$ gcc processId.c -o process
akhil@Akhil:~/OS$ ./process
Process ID=810
```

Program-4:

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
int main(){
fork();
fork();
printf("Hello\n");
return 0;
}
```

Output:

```
akhil@Akhil:~/OS$ gcc fork.c -o fork
akhil@Akhil:~/OS$ ./fork
Hello
Hello
Hello
```

Program-5:

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>

int main()
{
    pid_t pid;
    pid=fork();
    if(pid==0)
    {
        printf("I am Child Process\n");
    }
    else{
        printf("I am Parent Process\n");
    }
    return 0;
}
```

Output:

```
akhil@Akhil:~$ gcc parent.c -o parent
akhil@Akhil:~$ ./parent
I am Parent Process
I am Child Process
```

Program-6:

```
#include<stdio.h>
#include<unistd.h>
int main(){
pid_t pid;
pid=fork();
if(pid==0){
printf("Child PID=%d\n",getpid());
printf("Parent PID=%d\n",getppid());
}
else{
printf("Parent PID=%d\n",getpid());
}
return 0;
}
```

Output:

```
akhil@Akhil:~$ gcc parent1.c -o parent1
akhil@Akhil:~$ ./parent1
Parent PID=1028
Child PID=1029
Parent PID=927
```

Program-7:

```
GNU nano 8.7.1
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>

int main(){
    pid_t pid;
    pid=fork();
    if(pid==0)
    {
        printf("Child Running\n");
    }
    else{
        wait(NULL);
        printf("Parent Running After Child\n");
    }
    return 0;
}
```

Output:

```
akhil@Akhil:~$ ./wait
Child Running
Parent Running After Child
```